Intro to the Cluster

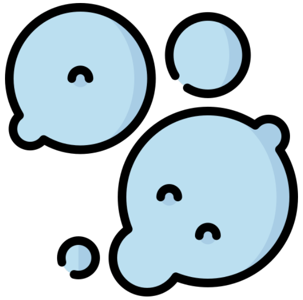
LaBella Lab
Wild Yeast Project

UNIVERSITY OF NORTH CAROLINA CHARLOTTE

Outline

- Brief summary of ONT (i.e., Nanopore)
- Intro to Practice Activity
- Practice Activity

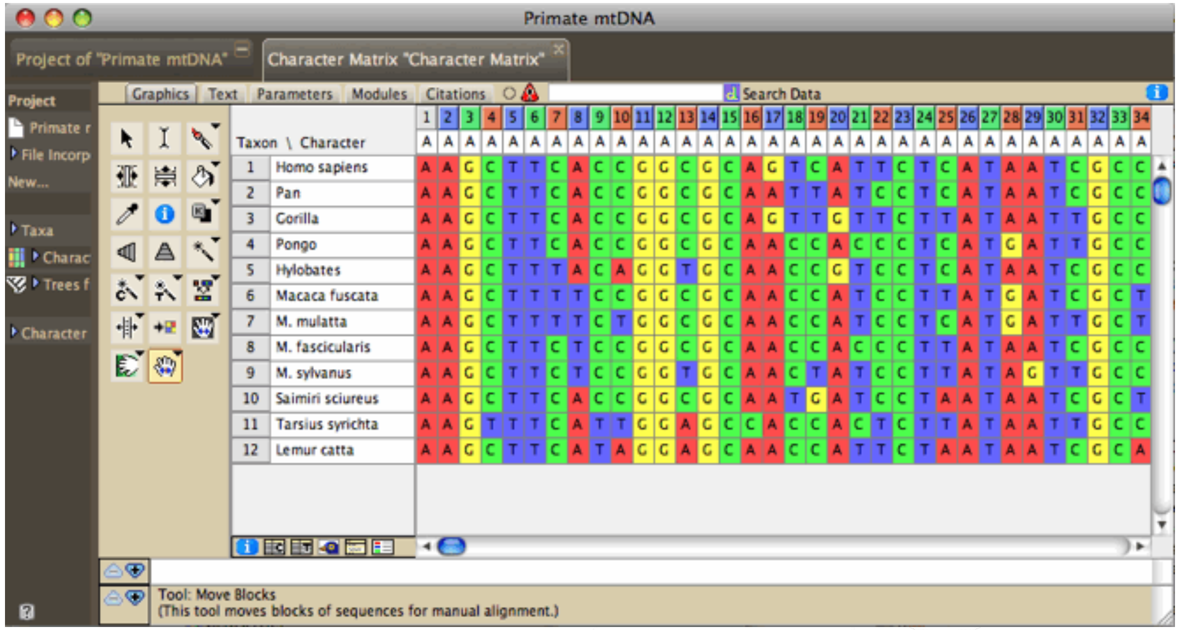
DNA Sequencing



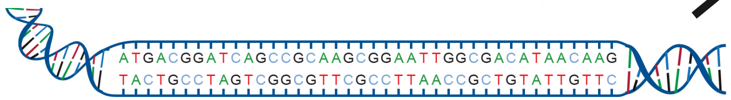
Yeast

DNA Extraction

Purified DNA =
Many copies of the
same sequence(s)

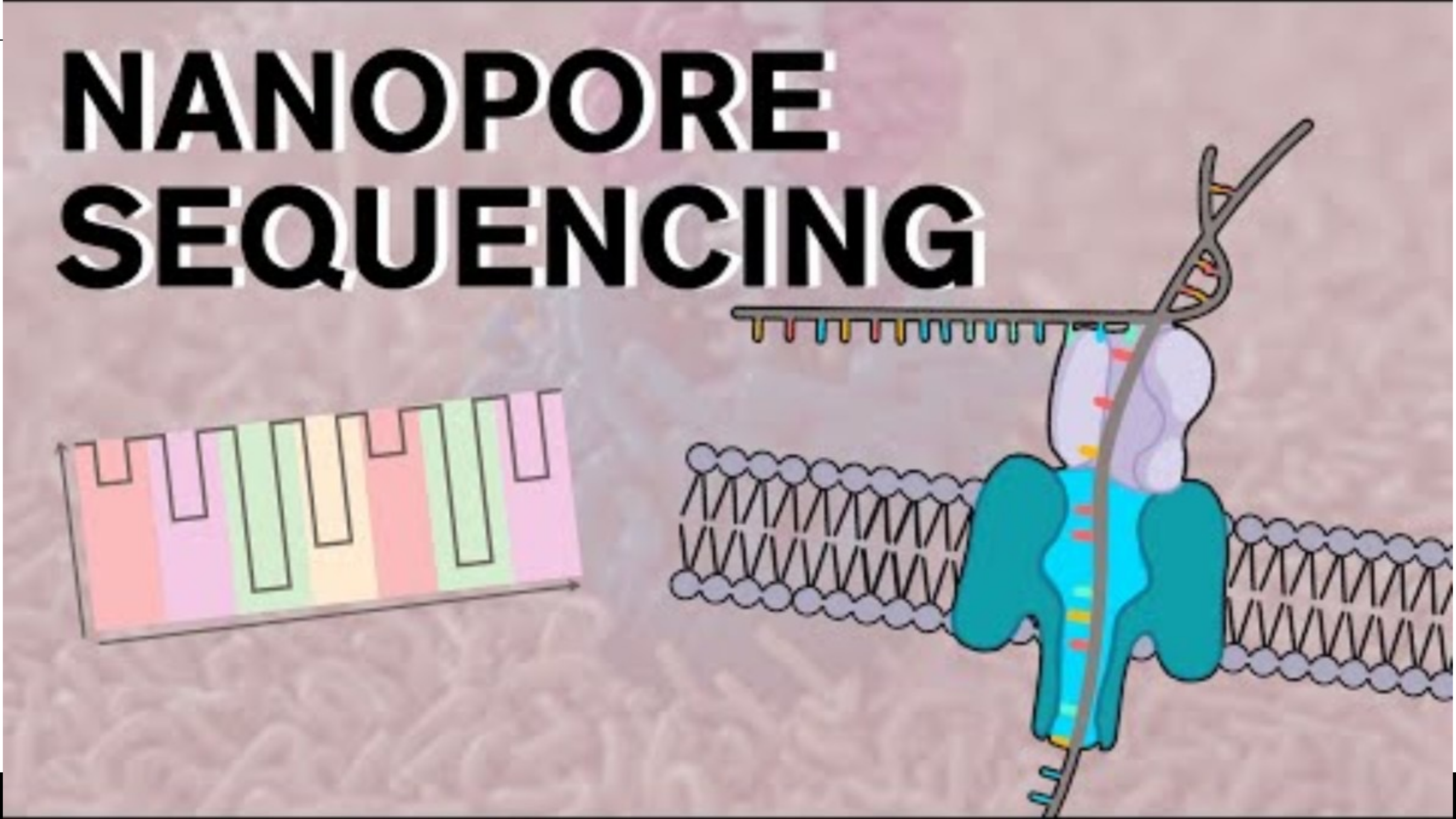


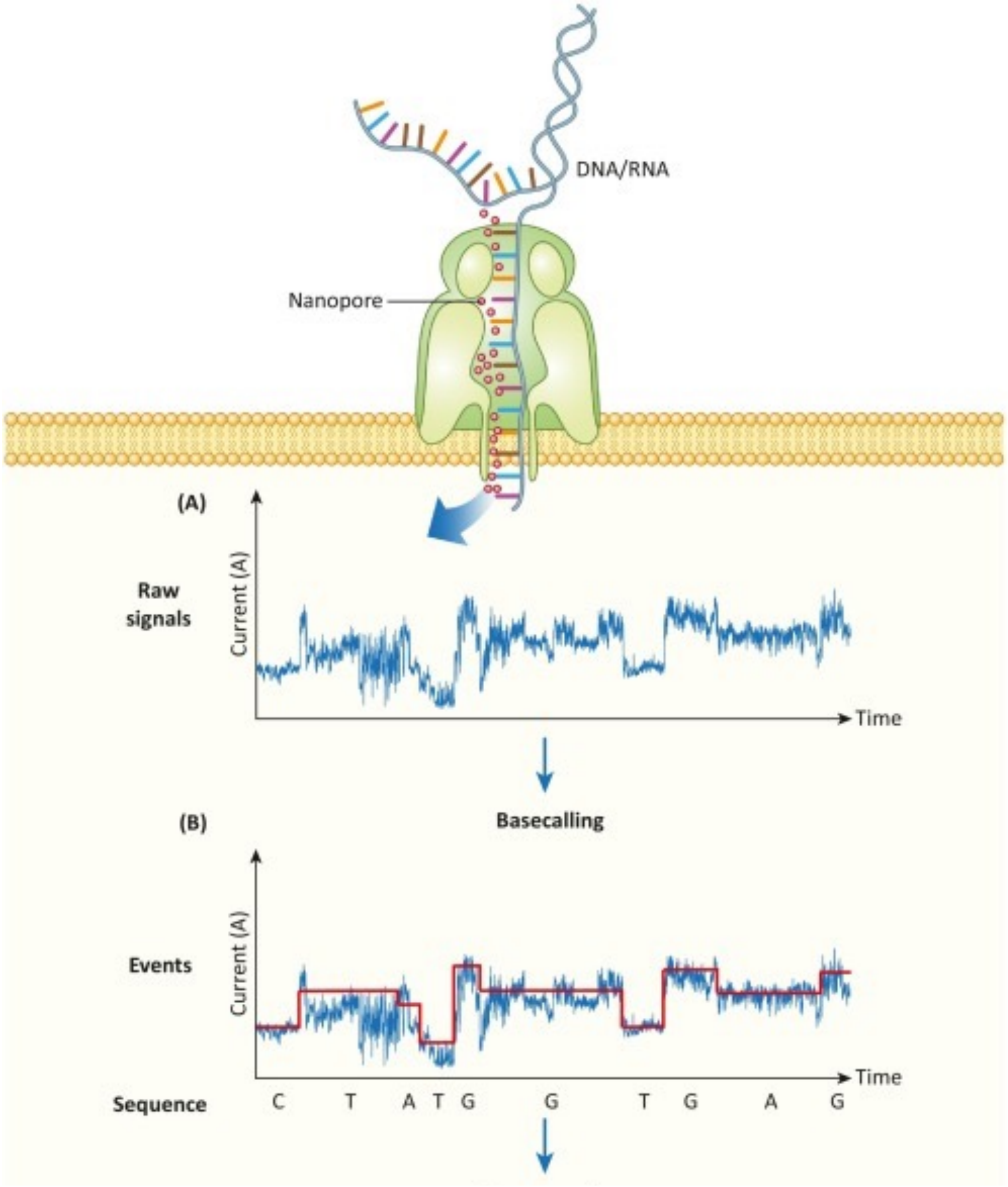
DNA Sequencing



Oxford Nanopore Sequencing

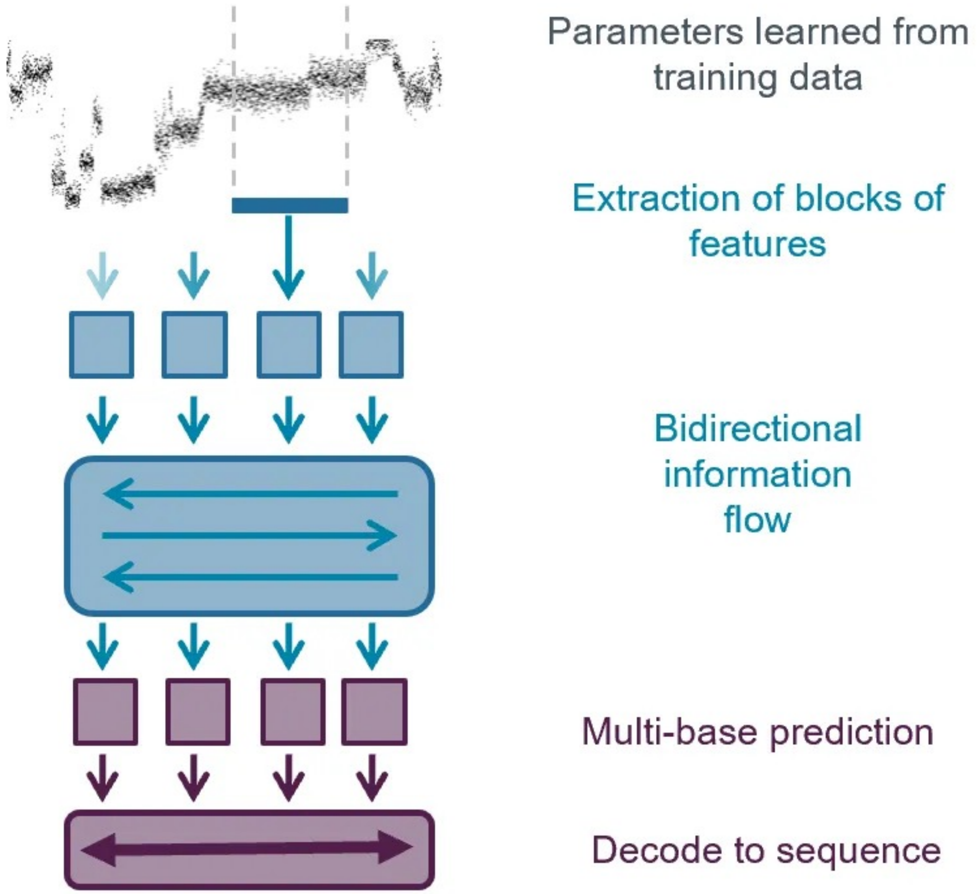
- Completely novel technology that does not require polymerase
- Does not sequence by synthesizing a new strand of DNA
- Holds the world record for longest DNA sequence (2.3 M bp)
- High (but not highest) accuracy


NANOPORE
SEQUENCING

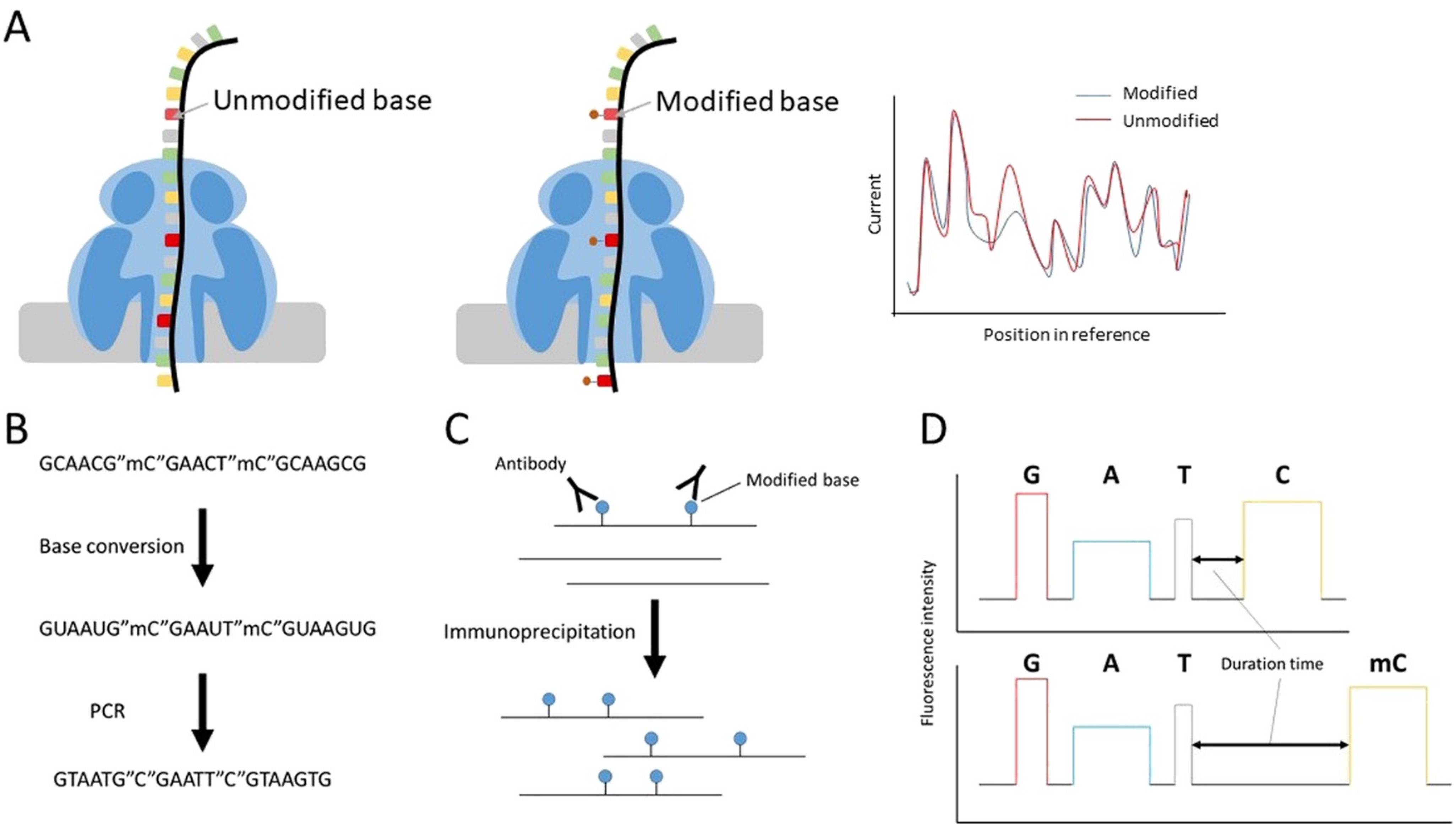

DNA/RNA
Nanopore
(A)
Raw signals
Current (A)
Time
Basecalling
(B)
Events
Current (A)
Time
Sequence
C T A T G G T G A G

Base Calling – Neural Networks

- Machine learning model trained on known data
- New models are constantly being developed



Detects modified bases



Practice Activity

https://github.com/The-Lab-LaBella/Wild_Yeast_Summer_2026/blob/main/Intro_to_Cluster.md